

CODE OF PRIORITY QUEUE

```
#include <iostream>

#include <queue>

#include <string>

using namespace std;

struct EmergencyVehicle
{
    string name;

    string type;

    int priority;

    bool operator<(const EmergencyVehicle &e) const
    {
        return priority > e.priority;
    }
};

int main()
{
    priority_queue<EmergencyVehicle> pq;

    int n;

    cout << "URBAN EMERGENCY PRE-EMPTION ENGINE\n";

    cout << "\nEnter number of emergency vehicles: ";

    cin >> n;

    for (int i = 0; i < n; i++)
    {
        EmergencyVehicle e;

        cout << "\nEmergency Vehicle " << i + 1 << endl;

        cout << "Enter vehicle name: ";

        cin >> e.name;

        cout << "Enter vehicle type: ";

        cin >> e.type;
```

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```
    cout << "Enter priority (1 = highest): ";
    cin >> e.priority;
    pq.push(e);
}
cout << "\nEmergency vehicles will be handled in this order:\n";
while (!pq.empty())
{
    EmergencyVehicle e = pq.top();
    pq.pop();
    cout << "\nVehicle: " << e.name;
    cout << "\nType: " << e.type;
    cout << "\nPriority: " << e.priority << endl;
    cout << "Status: Emergency request processed\n";
}
return 0;
}
```